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# Production Deployment & Technology Strategy

Ship lean, run it as an AI-operated money app, harden into a fraud-resistant, compliance-ready platform — grounded in the existing Argus Financial Partners architecture.

ARGUS FINANCIAL PARTNERS · CONFIDENTIAL — FOUNDER · JUNE 11, 2026

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# Production Deployment & Technology Strategy

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How to ship Argus Financial Partners to real users cheaply, run it as an AI-operated bank, and harden it into a fraud-resistant, compliance-ready platform — grounded in what's already built in the repo (Hedera settlement, the double-entry ledger, DID/VC + OID4VP + MCP, the risk-adaptive onboarding orchestrator, and the Phase 15 internal-agent-ops design).

**How to read this.** Recommendations are decisive and tagged **STAGE 0** (lean launch), **STAGE 1** (scale), or **STAGE 2** (licensed) so the tech rollout lines up with the three-phase compliance ramp in the *Corporate Structure* document. Costs are 2025–2026 planning estimates.

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## 1 1. Cloud — recommendation & cost envelope

**Verdict:** don't put a hyperscaler bill on a pre-revenue product. **Start on a lean managed PaaS, designed so the Stage-1 move to a single hyperscaler is a deploy-target change, not a rewrite.**

### Stage 0 — lean launch (recommended now)

The app is a stateless Node/TS API + Postgres + a few managed data services. Run it for tens of dollars a month:

| Concern                                       | Pick                                                             | Why / cost                                                                                  |
|-----------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| API (Node/Express/TS)                         | <b>Render</b> or <b>Fly.io</b>                                   | Git-push deploys, autoscale-to-cheap, private networking. ~\$7–25/mo                        |
| Postgres (prod<br><code>DATABASE_URL</code> ) | <b>Neon</b> (serverless) or <b>Supabase</b>                      | Branching, scale-to-zero, backups. Free→\$20/mo. The code already abstracts SQLite↔Postgres |
| Graph (Identity Vault)                        | <b>Neo4j Aura</b>                                                | Managed, free tier → serverless. \$0→\$65/mo                                                |
| Cache / queue                                 | <b>Upstash Redis</b>                                             | Per-request pricing, ~\$0 at low volume                                                     |
| Secrets / KMS                                 | <b>Infisical</b> or the PaaS secret store → cloud KMS at Stage 1 | Closes the <code>private_key_hex</code> gap (see §2)                                        |
| Object storage                                | <b>Cloudflare R2</b>                                             | No egress fees                                                                              |
| CDN / WAF / DNS                               | <b>Cloudflare</b>                                                | Free WAF + DDoS in front of everything                                                      |

**Stage-0 monthly burn:** ~\$0–150 depending on traffic. That's the whole point — stay near zero until usage justifies more.

### Stage 1 — consolidate on one hyperscaler (when traffic/compliance demands)

When you need VPC isolation, SOC 2 scope, managed Temporal/Conductor, and a data/ML platform, pick **one** cloud and commit:

|                          | <i>(recommended default)</i>                                | <i>(pick if ML/graph dominates)</i>                |
|--------------------------|-------------------------------------------------------------|----------------------------------------------------|
| Compliance ecosystem     | Broadest; most BaaS banks, custodians, KYC vendors run here | Strong, slightly thinner fintech-partner set       |
| Managed Postgres         | RDS / Aurora                                                | Cloud SQL / AlloyDB (excellent)                    |
| Containers/orchestration | EKS / ECS Fargate (host Temporal + Conductor)               | GKE Autopilot (cheapest managed k8s)               |
| ML / fraud               | SageMaker                                                   | <b>Vertex AI</b> (best-in-class, cheaper to start) |
| KMS / HSM                | KMS + CloudHSM (for the Hedera paymaster)                   | Cloud KMS + HSM                                    |
| Cost at scale            | Closeable with Savings Plans                                | Often 10–25% cheaper for data+ML                   |

**Recommendation:** default to **AWS at Stage 1** — a money business benefits most from AWS's compliance breadth and the fact that your future partner banks/custodians (Corporate doc, Phase B) overwhelmingly live there, easing private-link integrations and vendor SOC 2 alignment. Choose **GCP** only if the fraud-ML and Identity-Vault graph workloads become the cost center — Vertex AI + AlloyDB are a genuinely better value for that shape. **Do not split across both.**

## Rough cost envelope by scale

| Users   | Stage                 | Monthly infra (ballpark)                  |
|---------|-----------------------|-------------------------------------------|
| 0–1k    | Stage 0 (PaaS)        | <b>\$50–250</b>                           |
| ~1k–10k | Stage 0→1             | <b>\$300–1,500</b>                        |
| ~100k   | Stage 1 (AWS/GCP, HA) | <b>\$4k–12k</b> + data/ML + SOC 2 tooling |
|         |                       |                                           |
|         |                       |                                           |

## 2. Reference production architecture

The repo is already production-shaped (env-driven config that fails fast on insecure prod settings, Postgres-or-SQLite via `DATABASE_URL`, append-only ledger/audit triggers, RS256

token factory, pino + prom-client). Stage-1 wraps it in the standard hardening:

- **Network:** single VPC, private subnets for the DB/graph/Temporal; only the API + Cloudflare are public. mTLS / private link to partner-bank and custodian APIs at Phase B.
- **Secrets & keys — close the known gap.**  $\triangle$  **priority.** The repo documents that Phase 5 stores `private_key_hex` server-side (Hedera paymaster + any server-held keys). Move these to **cloud KMS/HSM**: envelope-encrypt at rest, never log (pino redaction already strips secrets/tokens/VC/VP), and for the Hedera **treasury/paymaster** use an HSM-backed key and a **multisig** so no single server compromise drains funds. User keys remain on-device (Secure Enclave) — never regress that.
- **CI/CD:** GitHub Actions → typecheck + the full vitest suite (131 pass / 3 todo) + the **Playwright web E2E** (21 tests) as a required gate → deploy. Migrations run forward-only (already idempotent).
- **Observability (already emitting):** scrape the five prom-client counters ( `ledger_post_total` , `vp_verify_total{result}` , `mcp_call_total{tool,result}` , `hedera_tx_total{result}` , `http_request_duration` ) into **Grafana Cloud** (free tier) with alerts; ship pino logs to a retained store; add request-id tracing (already threaded).
- **Ledger integrity:** the append-only `ledger_entries` / `ledger_journals` / `audit_logs` / `mcp_audit_logs` triggers are the spine — keep them; add **point-in-time-recovery** backups and a periodic **balanced-journal invariant** job (the invariant tests already encode the rule).
- **DR/backups:** automated PITR on Postgres + Neo4j, cross-region snapshot copies at Stage 1, a documented restore runbook, and quarterly restore drills.
- **Hedera mainnet cutover:** testnet → mainnet with the paymaster HSM/multisig above, plus the reconciliation job in §9 (the deferred on-chain↔ledger invariant `n` ).

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### 3 3. The automation fabric — agents, MCP & skills to run the bank

This is the differentiator: **an AI-operated bank** where, per the Phase 15 invariant already designed in `docs/PHASE-15-INTERNAL-AGENT-OPS.md` , **agents decide, deterministic code executes, and humans gate** anything irreversible. The runtime target is already locked: **Temporal** for money-critical workflows, **Conductor (OSS)** for agent orchestration. Build the catalog on the existing seam ( `signalService` , `riskOrchestratorService` , `orchestratorModel` , the `agent_runs` table, `tokenFactory` , `rbac` ).

## Two planes

- **External plane (built — Phase 7):** the MCP server lets *third-party* AI agents operate on a user's behalf under user-granted, VP-verified, scope-intersected, 90-second tokens. Already verified end-to-end (scoped-token mint, `SCOPE_DENIED`, balanced-journal transfer, `REPLAY_DETECTED`). This is a moat — keep investing here.
- **Internal plane (Phase 15 design → build):** operating agents that run the bank. Per the security model, internal skills are **read / recommend / draft only** — no agent gets direct money, state, regulator, or infra tools; a deterministic runner executes and a human gates.

## Internal operating-agent catalog

| Agent                              | Reads                                       | Produces<br>(draft/recommend)                    | Human gate                                            |
|------------------------------------|---------------------------------------------|--------------------------------------------------|-------------------------------------------------------|
| <b>Onboarding risk</b><br>(exists) | Signals,<br>device/email/IP/behavior        | Tier + risk decision<br>rationale                | Auto-approve low<br>risk; human on<br>review-required |
| <b>Compliance<br/>triage</b>       | Audit logs, sanctions hits,<br>txn patterns | SAR/CTR draft, case<br>summary                   | Compliance officer<br>files                           |
| <b>Dispute /<br/>chargeback</b>    | Ledger, MCP audit, comms                    | Resolution<br>recommendation + draft<br>response | Ops approves<br>payout/reversal                       |
| <b>Treasury ops</b>                | Balances, on-chain +<br>ledger              | Rebalance / paymaster<br>top-up proposal         | Human signs the<br>multisig                           |
| <b>Reconciliation</b>              | On-chain state vs ledger                    | Break report +<br>proposed adjusting<br>entry    | Human approves any<br>journal                         |
| <b>Support copilot</b>             | User state, docs, history                   | Drafted reply, next-<br>best-action              | Agent/human sends                                     |

## Skills (Claude Agent SDK / MCP tools)

Mirror the repo's own dogfooding pattern — it already ships operational skills ( `e2e-validator`, `argus-mcp-test-harness` ) that drive the system as a real client. Extend that idea to **ops skills** (reconciliation report, compliance-case summarizer, incident triage, release validator) so routine operations and CI are agent-automatable, while every state change still flows through the deterministic runner with its audit trail.

**Build sequence:** Conductor + the runner contract → onboarding-risk (already real) → reconciliation & compliance-triage (highest ops leverage) → treasury & dispute → support copilot.

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## 4 Identity Vault — a Neo4j graph of users & associations

A property graph is the right model for *relationships*, which is exactly where fraud and identity truth live. It complements (does not replace) the Postgres ledger.

**Core schema (nodes → relationships):**

```
(:User)-[:OWNS]->(:Device)           (:User)-[:USED]->(:IP)
(:User)-[:HOLDS]->(:Credential{VC})  (:User)-[:FUNDS_WITH]->(:Instrument)
(:User)-[:TRANSACTIONED_WITH]->(:User)  (:User)-[:NAMES]->(:Beneficiary)
(:User)-[:SHARES]->(:Attribute)        (:Wallet)-[:BOUND_TO]->(:DidKey)
```

**Fed from what already exists** — no new capture layer:

- onboarding **signals** (device fingerprint, email/IP/behavior scores) → device/IP/attribute edges;
- the **ledger** → **TRANSACTIONED\_WITH** edges with weights;
- **MCP audit logs** + VP/holder-binding → wallet↔DID↔credential edges;
- KYC/VC issuance → credential and shared-attribute edges.

**What it unlocks:** synthetic-identity rings (many users → one device/instrument), shared-device and shared-beneficiary fraud, collusion clusters, and "who is this account actually connected to" for compliance. Graph features (community, centrality, shortest-path-to-known-bad) feed the fraud model in §5.

**Hosting & guardrails:** Neo4j Aura (managed) at Stage 0/1 — don't self-host a database you'll stake fraud decisions on. Treat it as **PII**: field-level encryption for identifiers, strict access control, a documented **retention/erasure** policy (GDPR/CCPA right-to-delete must cascade), and **no raw PII in graph properties** you don't need — store hashes/refs where possible.

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## 5 5. Fraud detection — best-of-breed, built on your orchestration engine

The brief is right: **capture every good transaction and signal across the entire user lifecycle**, and learn normal so anomalies stand out. Build it *on top of the orchestration engine already in the product*, not as a bolt-on.

### Pipeline:

1. **Capture (lifecycle-wide)**: every onboarding signal, login, device, ledger journal, MCP call, marketplace action → an event stream. The append-only audit/ledger tables are the durable spine; add a streaming tap (Redis/Kafka-lite) for real-time.
2. **Feature store**: rolling user/device/graph features (velocity, novelty, peer-comparison, the Neo4j graph features from §4). Online (low-latency) + offline (training) views.
3. **Scoring (layered)**:
  - **Rules** (deterministic, explainable) — velocity caps, sanctions, impossible-travel, the existing >\$500 MFA gate and per-agent rate limits;
  - **ML anomaly** — unsupervised baseline (isolation forest / autoencoder) for "this is weird," plus a supervised model as labels accumulate;
  - **Graph signals** — proximity to known-bad clusters.
4. **Inline decision**: score **synchronously** on the money path (the ledger/MCP transfer flow already mints short-lived tokens and posts balanced journals — the natural choke point). Allow / step-up (MFA) / hold / block, each an auditable event.
5. **Feedback loop**: confirmed fraud/good labels flow back into the Identity Vault and retrain — the lifecycle-learning the brief describes.

**Build vs buy**: build the **orchestration + rules + graph layer in-house** (it's your moat and it sits on infra you own), and **buy the model/monitoring layer** if speed matters — **Unit21** or **Hawk AI** (AML/txn monitoring with case management) or **Sift** (consumer fraud).

Recommendation: in-house rules+graph from day one; add a vendor for AML case management at Phase B when the regulator expects it; keep the ML model in **Vertex AI/SageMaker** so you own the IP.

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## 6. Constellation Network "Digital Evidence" — a later-phase evaluation

**What it is:** Constellation's Hypergraph / Digital Evidence lets you notarize data and produce tamper-evident, independently-verifiable attestations of records and their lineage.

**Where it could fit:** as an *external anchor* over your already-append-only `audit_logs` / `mcp_audit_logs` / ledger — periodically committing cryptographic proofs so a regulator or counterparty can verify the audit trail wasn't altered, without trusting your database. It pairs naturally with the VC/DID evidence the product already issues.

| Pros                                                | Cons                                                       |
|-----------------------------------------------------|------------------------------------------------------------|
| Independent, tamper-evident compliance attestations | Adds a vendor + chain dependency                           |
| Strengthens regulator/auditor trust story           | Your DB triggers already give strong internal immutability |
| Aligns with the DID/VC, on-chain ethos              | Unproven ROI until a regulator/partner asks for it         |

**Recommendation: defer to Phase B/C (post-launch).** Internal append-only triggers + KMS + good backups are sufficient for launch. Revisit Constellation when a **partner bank, auditor, or regulator** specifically wants third-party-verifiable evidence — then it's a differentiator, not overhead. Track it; don't build it into v1.

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## 7. Mobile — launch iOS first

**Recommendation: iOS first, Android fast-follow.**

- The repo's **SwiftUI wallet (Phase 10)** is already the furthest-along native surface (Secure-Enclave P-256 signing key, VC-in-Keychain, `did:key` encoder matching the backend, OID4VP/OID4VCI deep links) — less new work to ship.
- **Secure Enclave** is the cleanest expression of the locked non-custodial key model and the passkey-first story; Android Keystore is equivalent but you've already invested in the iOS path.
- US-first demographics + higher iOS engagement/ARPU + App Store trust suit a finance app's first cohort.

- App Store review for crypto/finance is strict but predictable; lead with the non-custodial, no-customer-funds (Phase A) framing.

**Then Android:** the architecture is explicitly "native build, keys in Secure Enclave / **Android Keystore**" — the second platform is a known, planned port, not a redesign. (Note the documented gap: the iOS source is reviewed-but-unverified — it needs an Xcode build/sign/submit pass before any store submission.)

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## 8 8. Marketing & go-to-market

**Positioning (and the legal guardrail):** you are an **AI-operated, tokenization-first money app** — not "a bank." Honor the Corporate-doc naming rule: never imply you hold deposits or are chartered. Lead with the wedge that's actually novel: **AI agents that act on your money under cryptographic, user-granted permission, and tokenized real-world assets** you can hold in a non-custodial wallet.

- **Wedge:** the agent + MCP experience (let your AI assistant check balances, move money, and trade under scoped, revocable, audited permission) is a genuinely differentiated story in 2026.
  - **Early segment:** crypto-comfortable, AI-forward early adopters and RWA-curious investors — not the mass market. Invite-only **waitlist** with referral mechanics to manufacture scarcity and control onboarding risk.
  - **Compliance-safe messaging:** "tokenized assets," "non-custodial wallet," "agentic finance" — never "deposits," "FDIC," "bank account," or investment-return promises. Every investment surface carries the "not advice / not an offering" disclaimer the product already models.
  - **Content & community:** build in public on the agent/MCP + DID/VC tech; developer-credible content earns the early crypto/AI crowd. Position the MCP server as a platform third parties can build on.
  - **Narrative:** "the first bank run by AI agents — that you actually control." Differentiation, not feature parity.
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## 9. Tech gaps to close (additive checklist)

| #  | Gap                                     | Priority  | Note                                                                         |
|----|-----------------------------------------|-----------|------------------------------------------------------------------------------|
| 1  | <b>KMS / HSM for server-held keys</b>   | △ Stage 1 | The documented <code>private_key_hex</code> gap; treasury multisig           |
| 2  | <b>On-chain ↔ ledger reconciliation</b> | Stage 1   | The deferred invariant <code>n</code> ; powers the reconciliation agent (§3) |
| 3  | <b>Data warehouse + analytics</b>       | Stage 1   | BigQuery/Redshift; product + risk analytics                                  |
| 4  | <b>Feature store</b>                    | Stage 1   | Underpins §5 fraud ML (Feast or managed)                                     |
| 5  | <b>Model registry + monitoring</b>      | Stage 1   | Versioning, drift, explainability for fraud models                           |
| 6  | <b>Incident response + on-call</b>      | Stage 0→1 | Runbooks, paging, status comms                                               |
| 7  | <b>SOC 2 readiness automation</b>       | Phase B   | Vanta/Drata; partners will require SOC 2 Type II                             |
| 8  | <b>Public status page</b>               | Stage 0   | Trust signal; ~free                                                          |
| 9  | <b>WAF / DDoS / bot defense</b>         | Stage 0   | Cloudflare in front (already in the stack)                                   |
| 10 | <b>Secrets rotation + key ceremony</b>  | Stage 1   | Especially the Hedera treasury keys                                          |
| 11 | <b>Dedicated staging env</b>            | Stage 0   | Mirror of prod; the E2E suite already targets it                             |
| 12 | <b>Rate-limit hardening at the edge</b> | Stage 0   | The app limiter exists; add edge limits too                                  |
| 13 | <b>Backup restore drills</b>            | Stage 1   | Test DR quarterly, not just configure it                                     |

## 10. Phased production roadmap (aligned to the compliance ramp)

| Compliance phase<br>(Corp doc) | A (non-custodial)                  | B (partnered, MSB)                            | C (own licenses)                       |
|--------------------------------|------------------------------------|-----------------------------------------------|----------------------------------------|
| Cloud                          | Render/Fly + Neon + Aura           | AWS (or GCP) single VPC                       | Multi-AZ HA, audited                   |
| Keys                           | On-device + Infisical              | <b>KMS/HSM + treasury multisig</b>            | HSM key ceremonies                     |
| Agents                         | Onboarding-risk live; MCP external | + reconciliation, compliance-triage, treasury | Full AI-ops with regulator-grade audit |
| Fraud                          | Rules + graph (in-house)           | + ML anomaly + AML vendor                     | Tuned models + case management         |
| Identity Vault                 | Neo4j Aura, core schema            | Graph features into fraud model               | Regulator-facing link analysis         |
| Mobile                         | <b>ios</b> (verify Xcode build)    | + Android                                     | Both, hardened                         |
| Evidence                       | Append-only triggers + KMS         | Evaluate Constellation                        | Constellation if a partner needs it    |
| Compliance tooling             | ToS/AML policy + OFAC              | SOC 2 (Vanta/Drata)                           | Exams, MTL/transfer-agent              |

**The throughline:** every stage spends only what the prior stage's traction earns. You launch on ~\$100/mo of PaaS as a non-custodial software product; you bring KMS, ML fraud, a second mobile platform, and SOC 2 only when partners and scale require them; and you reserve the expensive hyperscaler + licensing posture for when revenue justifies it — the same lean-now / hardened-later logic that drives the corporate structure.

*Companion document: Corporate Structure & Compliance Strategy (the lean Wyoming-LLC → Delaware-C-corp path and the Phase A/B/C compliance ramp this roadmap aligns to).*